

Distinguishing Exponential from Linear Change

Deciding from a table whether change is by a constant difference or a constant ratio, writing the matching rule, comparing a linear model with an exponential one, and classifying real situations before computing

The one idea behind every problem. Two patterns look alike for a step or two and then part company forever. Linear change **adds** the same amount each step; exponential change **multiplies** by the same factor each step. Every problem here asks the same first question: is the step an addition or a multiplication?

Linear: constant *difference* $f(n) = a + d \cdot n$

Exponential: constant *ratio* $f(n) = a \cdot r^n$

a is the starting value, d is what is added, r is what you multiply by

Reference card. No values are filled in — use the numbers given in each problem.

Directions.

- For every table, write the difference row *and* the ratio row before deciding. One of them will be constant, or neither will be.
- Say which model you chose and why, then compute.
- Round money answers to the nearest cent. Write your final answer on the answer line.

1. Here is a table of values for f .

n	0	1	2	3
$f(n)$	5	8	11	14

- (a) Write the differences between consecutive values, and the ratios. Which one is constant?
- (b) Is f linear or exponential? Write its rule.
- (c) Find $f(6)$.

Answer: _____

2. Here is a table of values for g .

n	0	1	2	3
$g(n)$	4	12	36	108

- (a) The differences here are 8, 24, 72 — not constant. Compute the ratios instead. What do you find?
- (b) Is g linear or exponential? Write its rule.
- (c) Find $g(5)$.

Answer: _____

3. Rosa opens a savings account with 200 dollars and adds 25 dollars at the end of every month. No interest is paid.

- (a) Is the balance changing by a constant difference or a constant ratio? Say which, in one sentence.
- (b) Write the rule for the balance after n months.
- (c) How many dollars are in the account after 8 months?

Answer: _____

4. A bacteria colony starts with 6 cells, and the number of cells doubles every hour.
- (a) “Doubles” names an operation. Is it addition or multiplication? What does that make the model?
 - (b) Write the rule for the number of cells after n hours.
 - (c) How many cells are there after 7 hours?

Answer: _____

5. Two savings plans both start at 100 dollars.
- **Plan L** adds 50 dollars each step.
 - **Plan E** multiplies by 1.5 each step.
- (a) Write a rule for each plan.
 - (b) Plan E is behind Plan L at step 1. Explain in one sentence how it can still end up ahead.
 - (c) At step 5, how many dollars is Plan E ahead of Plan L? Round to the nearest cent.

Answer: _____

6. Not every pattern is one of the two. Here is a table for h .

n	1	2	3	4
$h(n)$	2	8	18	32

- (a) Compute the first ratio $h(2) \div h(1)$.
(b) Compute the next ratio $h(3) \div h(2)$ as a decimal.
(c) The differences are 6, 10, 14 — not constant either. What do your two ratios prove about h ? Write both ratios on the answer line.

Answer: _____

7. A pattern is known to be linear, and two of its values are $f(2) = 17$ and $f(5) = 32$.
(a) What is the constant rate of change?
(b) Use it to write the rule for $f(n)$.
(c) If instead the pattern were exponential through those same two points, would the rate of change still be constant? Answer in one sentence.

Answer: _____

8. A car is worth 24000 dollars today. Two dealers describe how it will lose value.

- **Dealer A:** it loses 12 percent of its current value each year.
- **Dealer B:** it loses 2400 dollars each year.

(a) Which description is exponential and which is linear? How do you know?

(b) Write a rule for each value after n years.

(c) After 5 years, how many dollars more is the car worth under Dealer A than under Dealer B? Round to the nearest cent.

Answer: _____

9. A population grows exponentially. It is 500 at time 0 and 4000 at time 3.

(a) Find the growth factor r per unit of time. Show the equation you solved.

(b) Use your rule to find the population at time 5.

(c) Write both answers on the answer line, in the order (a) then (b).

Answer: _____

10. Challenge. A summer job offers two pay plans for a 20-day stretch.

- **Flat plan:** 1000 dollars on every day.
- **Doubling plan:** 0.01 dollars on day 1, and the daily pay doubles each day after that.

(a) Write a rule for the doubling plan's pay on day n . Be careful with the exponent: day 1 has been doubled zero times.

(b) On day 10, which plan pays more that day? A one-line estimate is enough.

(c) On day 20, how many dollars more does the doubling plan pay that day than the flat plan? Round to the nearest cent.

Answer: _____